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The 1420 Ma Indiavaí Mafic Intrusion (SW Amazonian Craton): Paleomagnetic results and implications for the Columbia supercontinent

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ABSTRACT

The configuration and the timing of assembly and break-up of Columbia are still matter of debate. In order to improve our knowledge about the Mesoproterozoic evolution of Columbia, a paleomagnetic study was carried out on the 1420 Ma Indiavaí mafic intrusive rocks that crosscut the polycyclic Proterozoic basement of the SW Amazonian Craton, in southwestern Mato Grosso State (Brazil). Alternating field and thermal demagnetization revealed south/southwest ChRM directions with downward inclinations for sixteen analyzed sites. These directions are probably carried by SD/PSD magnetite with high coercivities and high unblocking temperatures as indicated by additional rock magnetic tests, including thermomagnetic data, hysteresis data and the progressive acquisition of isothermal remanent magnetization. Different stable magnetization components isolated in host rocks from the basement 10 km NW away to the Indiavaí intrusion, further support the primary origin of the ChRM. A mean of the site mean directions was calculated at Dm = 209.8°, Im = 50.7° (α_{95} = 8.0°, K = 22.1), which yielded a paleomagnetic pole located at 249.7°E, 57.0°S (A_{95} = 8.6°). The similarity of this pole with the recently published 1420 Ma pole from the Nova Guarita dykes in northern Mato Grosso State suggests a similar tectonic framework for these two sites located 600 km apart, implying the bulk rigidity of the Rondonian-San Ignacio crust at that time. Furthermore these data provide new insights on the tectonic significance of the 1100–1000 Ma Nova Brasilândia belt—a major EW feature that cuts across the basement rocks of this province, which can now be interpreted as intracratonic, in contrast to previous interpretation. From a global perspective, a new Mesoproterozoic paleogeography of Columbia has been proposed based on comparison of these 1420 Ma poles and a 1780 Ma pole from Amazonia with other paleomagnetic poles of similar age from Baltica and Laurentia, a reconstruction in agreement with geological correlations.

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1. Introduction

The existence of a Paleo- to Mesoproterozoic supercontinent has been postulated by various authors, although its configuration is still a matter of debate, partly due to the scarcity of paleomagnetic data (Rogers, 1996; Meert, 2002; Rogers and Santosh, 2002; Zhao et al., 2002, 2004, 2006; Pesonen et al., 2003; Kusky et al., 2007; Bispo-Santos et al., 2008; Hou et al., 2008; Johansson, 2009; Evans and Mitchell, 2011). Several names are attributed to this supercontinent—Nena, Nuna, Paleopangea, Columbia, among others (Gower et al., 1990; Hoffman, 1997; Piper, 2000; Rogers and Santosh, 2002). Following the reasoning of Meert (2012) we call this supercontinent as Columbia.

In the present work we focus on some of the building blocks of Columbia, namely Baltica, Laurentia, Amazonia, North China and West Africa. Two different configurations for the Laurentia–Baltica connection have been proposed in Columbia supercontinent, one of which juxtaposes southeastern Greenland to western Baltica (Zhao et al., 2002, 2004; Hou et al., 2008; Johansson, 2009), and the other one places eastern Greenland next to northern Baltica (Hoffman, 1988; Buchan et al., 2000; Karlstrom et al., 2001; Pesonen et al., 2003; Zhao et al., 2006; Salminen and Pesonen, 2007; Bispo-Santos et al., 2008; Lubnina et al., 2010; Pisarevsky and Bylund, 2010; Evans and Mitchell, 2011). Paleomagnetic data favor the latter model, suggesting Laurentia and Baltica drifted together since 1800 Ma up to at least 1270 Ma ago (Salminen and Pesonen, 2007; Lubnina et al., 2010; Pisarevsky and Bylund, 2010; Evans and Mitchell,

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2011). Geologic and geochronologic similarities in the Mesoproterozoic evolution of Laurentia and Baltica with that of the Amazonian Craton led several authors to propose a possible link between the north-northeastern Amazonian Craton and southwestern Baltica (e.g., Sadowski and Bettencourt, 1996; Geraldes et al., 2001; Zhao et al., 2002, 2004, 2006; Hou et al., 2008; Johansson, 2009). Both cratons show subduction-related accretionary belts, which evolved throughout the Mesoproterozoic marginally to such contiguous cratonic nuclei. However, paleomagnetic tests of this configuration for Paleoproterozoic times (Pesonen et al., 2003; Bispo-Santos et al., 2008) indicate that these cratons were not linked. Instead, a well-dated 1789 ± 7 Ma (SHRIMP U–Pb in zircon) paleomagnetic pole from the Amazonian Craton (the Colider intrusive suite) suggested a latitudinal gap between this craton and the Laurentia–Baltica block, which could have been occupied by another large landmass, such as the North China Craton (Bispo-Santos et al., 2008). The position of North China between Baltica and Amazonia is supported by the models of Kusky et al. (2007) and Yakubchuk (2010), which were based on the continuity of Archean and Paleoproterozoic provinces along those four different cratons.

Recently, Bispo-Santos et al. (2012) reported new paleomagnetic and ^{40}Ar–^{39}Ar results for the 1418.5 ± 3.5 Ma Nova Guarita dyke swarm from northern Mato Grosso State (southwestern Amazonian Craton). These new data provided a test for the longevity of the previously proposed Columbia paleogeography. The mismatch between Laurentia–Baltica and Amazonian poles led the authors to suggest that break-up of Columbia must have taken place before 1420 Ma, although a paleogeography where Amazonian Craton was closer to Baltica in the Mesoproterozoic is also allowed by the paleomagnetic data. In the present study, we report new paleomagnetic data from the recently dated (1415.9 ± 6.9; Teixeira et al., 2011) Indiavai mafic intrusion sampled in southwestern Mato Grosso State, ca. 600 km away from the Nova Guarita region. We use the new paleomagnetic data to test different paleogeographic configurations of Columbia for the Mesoproterozoic (and also Paleoproterozoic) times, and speculate on the viability of the different paleogeographic configurations proposed so far, given the present paleomagnetic and geological evidence.

2. Geological setting

The Amazonian Craton is one of the largest tectonically stable areas of Precambrian rocks in the world. The northeastern part of the craton consists of two Archean blocks with granite-greenstone terranes and high-grade metamorphic rocks exposed in the Central Amazonian Province (Fig. 1a). These Archean blocks collided along the Maroni-Itacaiunas belt, between 2250 and 2050 Ma ago (e.g., Ledru et al., 1994; Santos et al., 2000; Fraga et al., 2009). The basement rocks are covered by undeformed volcano-sedimentary successions dated between 1900 and 1400 Ma, and are bordered by progressively younger accretionary belts towards the southwest (Tassinari et al., 2000; Cordani and Teixeira, 2007; Cordani et al., 2009; Bettencourt et al., 2010; Teixeira et al., 2010) (Fig. 1a): the 1980–1810 Ma Ventuari-Tapajós Province, the 1780–1600 Ma Rio Negro-Juruena Province, the 1600–1300 Ma Rondonian-San Ignacio Province, and the 1200–950 Ma Sunsás-Aguapeí Province. The latter includes the Nova Brasilândia high grade belt (see afterward).

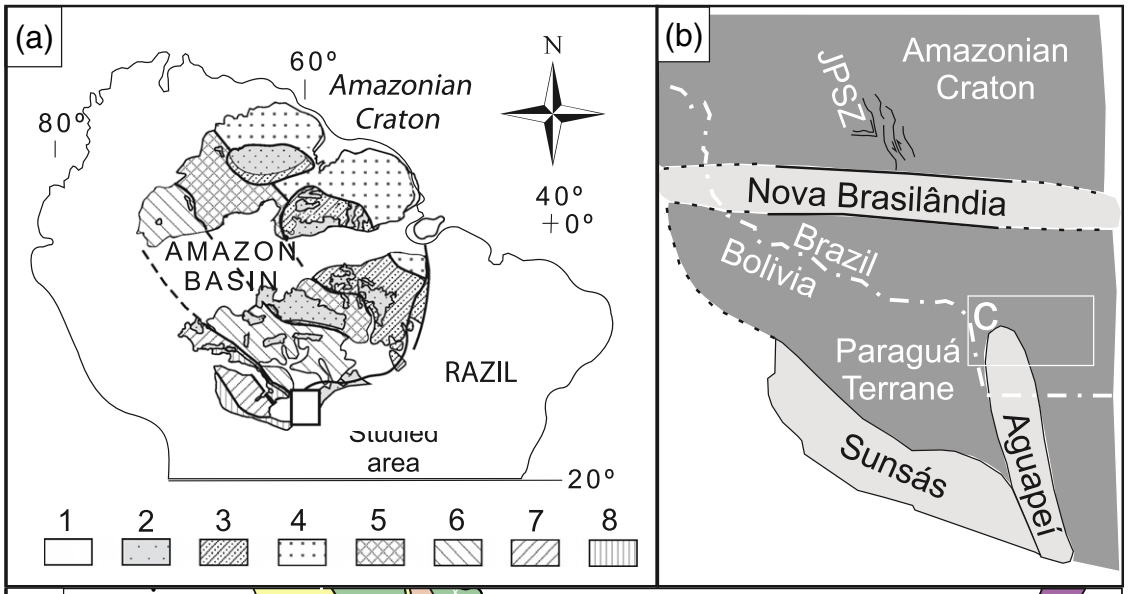
The southwestern border of the Amazonian Craton has experienced a complex geological history involving the Neoproterozoic Sunsás, Aguapeí and Nova Brasilândia belts and the Paleo to Mesoproterozoic Paraguá Terrane (Fig. 1b)—see Teixeira et al. (2010) for a review. The Sunsás belt, which marks the southwestern border of the Craton, has long been considered the counterpart of the collisional Grenville belt in Rodinia reconstructions (e.g., Dalziel, 1991; Hoffman, 1991; Sadowski and Bettencourt, 1996). The intracratonic Aguapeí basin (Fig. 1b) is interpreted as an aborted rift (Saes and Fragoso-Cesar, 1994), being separated from the Sunsás belt sensu strictu by the Paraguá Terrane, a Paleo and Mesoproterozoic basement block not affected by the Sunsás orogeny. According to Teixeira et al. (2010) the Aguapeí belt (or aulacogen) may be a late tectonic offshoot of the Sunsás collision over the Jauru granite-greenstone terrane. The northern limit of the Paraguá Terrane, is defined by the E–W 1100–1000 Ma Nova Brasilândia metasedimentary belt (Fig. 1b), which has been interpreted as the Meso- to Neoproterozoic suture zone between this Terrane and the Amazonian Craton basement to the north (Tohver et al., 2004, 2005a). However, this interpretation is disputed by Teixeira et al. (2006), Cordani and Teixeira (2007), and Cordani et al. (2009) who argue that deformation along the Nova Brasilândia belt rather represents intracratonic tectonic reactivation as result from the Sunsás collisional orogeny.

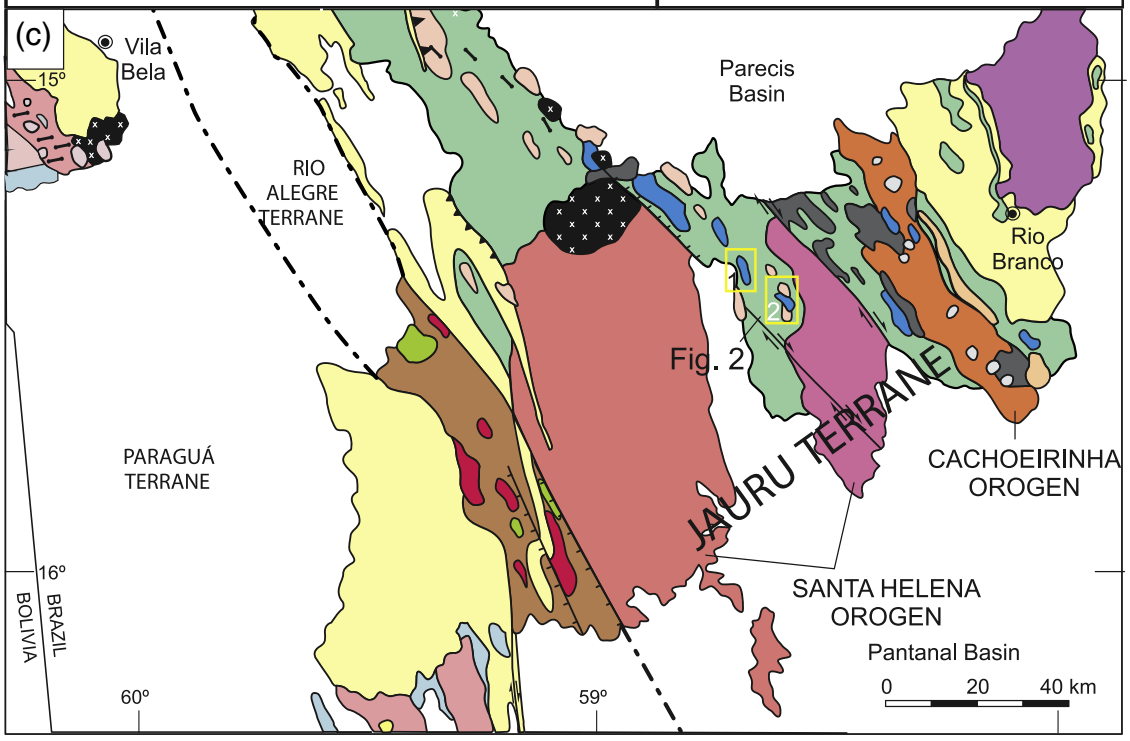
The focus of this work is an area situated to the southeast of the Nova Brasilândia belt (C rectangle in Fig. 1b), within the Jauru Terrane (1780–1420 Ga), which is interpreted by Bettencourt et al. (2010) as a SE extension of the Rio Negro-Juruena Province, whilst reworked by the mobile belts (Cachoeirinha, Santa Helena, Rio Alegre—see below) that built up the Rondonian-San Ignacio Province. In brief, the NW-trending Jauru Terrane (Fig. 1c) is represented by: (i) Paleoproterozoic basement rocks (1790–1720 Ma) comprising the Alto Jauru Group (metasedimentary–metavolcanic assemblages; 1760–1720 Ma), the Alto Guaporé Metamorphic Complex (mainly granite-gneisses; 1790–1740 Ma), and the Cabaçal Tonalite (1780 Ma) (Fig. 1); (ii) rocks from the Cachoeirinha accretionary orogen (1560–1520 Ma; Santa Cruz and Alvorada Intrusive Suites), composed of tonalites, granodiorites and granites; and (iii) younger intrusive associations (Rio Branco, Pindaituba, Santa Helena, and Água Clara), comprising tonalites, granites and mafic rocks making up the Santa Helena orogen (1.48–1.42 Ga, Bettencourt et al., 2010).

The Jauru Terrane is limited to the west by the Rio Alegre Terrane (1510–1380 Ma), (Saes and Fragoso Cesar, 1996). After uplift and cooling, these rocks were probably sources to the younger sedimentary rocks that formed the Sunsás-Aguapeí province (1200–950 Ma, Teixeira et al., 2010). Detrital zircon U–Pb ages from sedimentary rocks of the Fortuna Formation (lowest unit of the Aguapeí Group) are between 1500 Ma and 1230 Ma (Santos et al., 2001; Leite and Saes, 2003). This indicates a maximum age of 1230 Ma for deposition of the Fortuna Formation, whose possible sources are derived from the Rio Alegre Terrane and the granitoid suites that crosscut the Jauru Terrane (Leite and Saes, 2003).

The Figueira Branca Intrusive Suite (FBIS) consists of differentiated, disrupted mafic–ultramafic lithotypes comprising dunites, anorthosites, troctolites, norites, and gabbros. The latter rocks represent the upper part of one stratiform complex. From the petrographic point of view the gabbros may exhibit adcumulatic and cumulatic textures. The FBIS (Fig. 1c) is formed by several bodies mainly intruded into the Orosirian granite-gneisses and the metavolcano-sedimentary rocks of the Alto Jauru Group (Lacerda-Filho et al., 2004). One of the mafic lithotypes (rectangle #2 in Fig. 1c), the Indiavaí mafic intrusion crops out in the vicinity of the Indiavaí town, and another occurs along the Jauru River, in the Figueira Branca farm (rectangle #1 in Fig. 1c). The Indiavaí Intrusion represents a nearly elliptical pluton trending N20–30W (Fig. 2). It intrudes meta-volcanosedimentary rocks of the Alto Jauru Group, it is composed of melanocratic and mesocratic, medium-grained gabbroic rocks with intergranular, cumulate texture, and it presents a discrete tectonic orientation parallel to the main trend (N20–30W) of the pluton. In the studied area, it appears in close association with granitoid stocks, plugs and plutons (Pindaituba Intrusive Suite; 1462–1423 Ma), and show a similar tectonic strike (Ruiz, 2005).

Recently, Teixeira et al. (2011) reported ^{40}Ar–^{39}Ar and U–Pb (SHRIMP) geochronological data for gabbros from both intrusive occurrences. U–Pb (SHRIMP) zircon ages of 1415.9 ± 6.9 Ma and 1425.5 ± 8.0 Ma, from the Indiavaí and Figueira Branca intrusions, respectively, were interpreted as crystallization ages. The analytical concordance between these two sets of results indicates that they reflect a single magmatic event. Nevertheless, the ^{40}Ar–^{39}Ar dating





MESOZOIC-CENOZOIC
Undifferentiated sedimentary covers

NEOPROTEROZOIC
Aguapeí Group
Guapé Intrusive Suite

PALEO-MESOPROTEROZOIC

PARAGUÁ TERRANE
SAN IGNACIO OROGEN (1.37-1.31 Ga)
Late-to-post kinematic granites
Syn-to-late kinematic granites
Pensamiento Granitoid Complex
BASEMENT ROCKS
Chiquitania Complex

RIO ALEGRE TERRANE
RIO ALEGRE OROGEN (1.51-1.38 Ga)
Santa Rita Suite (Syn-to late kinematic granites 1.44-1.38 Ga.)
Mafic-ultramafic Intrusive Suite (1.51-1.49 Ga)
Rio Alegre volcanic-sedimentary unit (1.51 Ga).

JAURU TERRANE
Figueira Branca Intrusive Suite-FIBS (1.42 Ga) → 1. Figueira Branca 2. Indiavaí

SANTA HELENA OROGEN
Rio Branco Intrusive Suite
Pindaituba Intrusive Suite
Santa Helena Intrusive Suite
Água Clara Intrusive Suite
1.46-1.42 Ga

CACHOEIRINHA OROGEN
Alvorada Intrusive Suite
Santa Cruz Intrusive Suite
1.56-1.52 Ga

PALEOPROTEROZOIC BASEMENT
Cabaçal tonalite (1.78 Ga)
Alto Guaporé metamorphic Complex (1.79-1.74 Ga)
Alto Jauru Group (1.76-1.72 Ga)

Inferred tectonic boundary
Normal fault
Upright normal fault
Thrust fault
Strike-slip faults with shear sense
Geological contact
Dyke

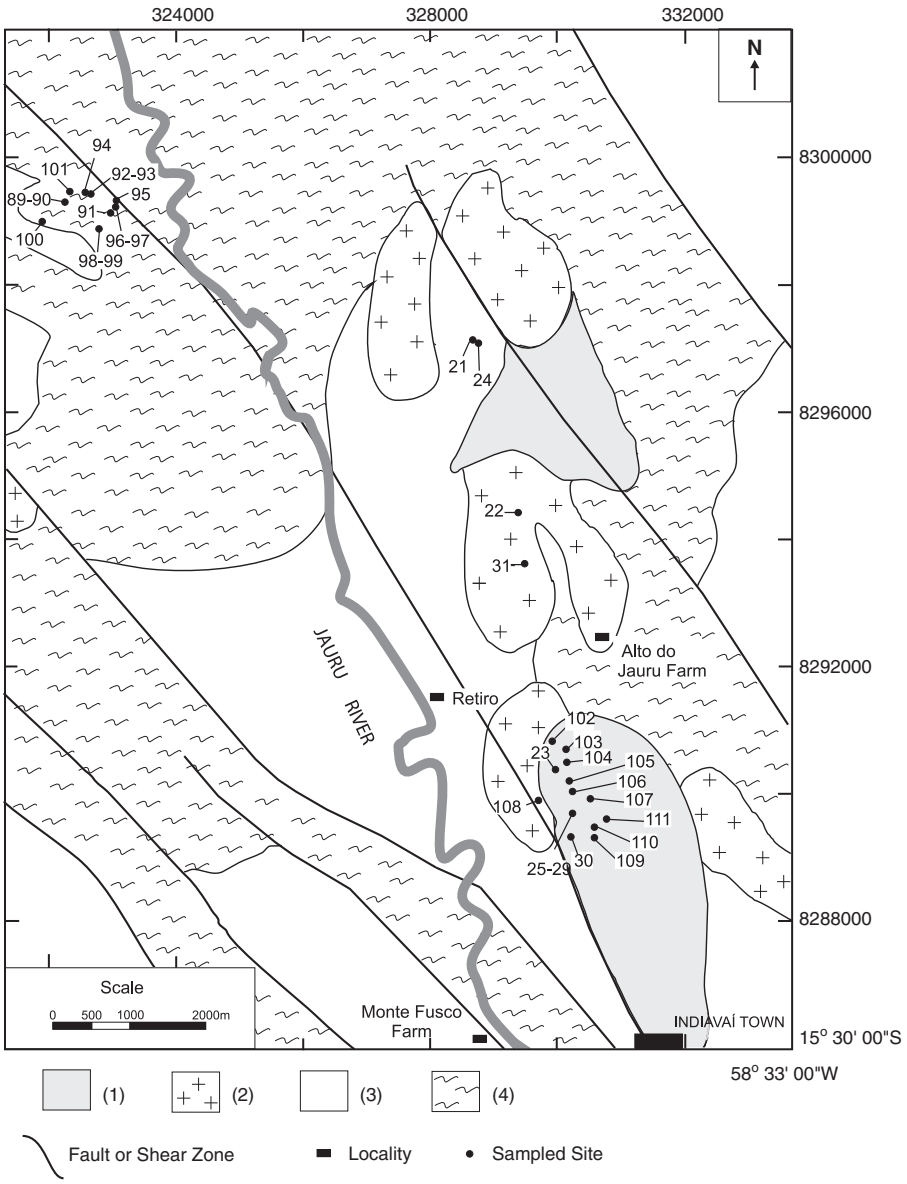


Fig. 2. Simplified geological map showing the Indiavaí mafic body and basement rocks (adapted from Nunes et al., 2000; Bettencourt et al., 2010; Teixeira et al., 2011). Indiavaí mafic intrusion (1420 Ma) (1); Pindaituba Intrusive Suite (1460–1420 Ma) (2); migmatites and gneisses (3) and metavolcanic–sedimentary sequence (4) of the Alto Jauru Group (1760–1720 Ma). Filled circles represent sampling sites (numbers as in Tables 1 and 2).

on biotite separated from Indiavaí gabbros yielded much younger plateau ages of 1222 ± 3 Ma and 1222 ± 4 Ma (ideogram age of 1222 ± 5 Ma), and apparent ages of 1275 ± 4 Ma and 1268 ± 4 Ma (Teixeira et al., 2011). This suggests that either these rocks were affected by a younger tectonothermal event or that they reflect very slow regional uplift and cooling (e.g., Tohver et al., 2006; Elming et al., 2009a)—see further in discussion.

3. Sampling and methods

Six to nine oriented drill cores were taken from 16 sites from pristine gabbros of the Indiavaí intrusion located to the north of the town of Indiavaí (Fig. 2). The sites are generally represented by in situ rock exposures that crop out along the pastures of local farms. In addition, for a baked contact test 42 oriented drill cores were taken from 5 sites (AZ-21, AZ-22, AZ-24, AZ31, AZ108), represented by felsic and gneissic basement country rocks located close to the Indiavaí Intrusion (Fig. 2). Another 70 oriented drill cores were taken from 13 sites (AZ-89 to AZ-101 in Fig. 2) in the metabasic basement country rocks (Alto Jauru Group) far from the Indiavaí Intrusion. These outcrops are located to the west of the Jauru River, northwest of the Indiavaí sites, and were collected along roads and streambeds where they are well exposed (Fig. 2). Cylindrical cores were drilled using a portable, gasoline-powered rock drill. Both solar and magnetic compasses were used, but

Fig. 1. (a) Amazonian Craton and their main tectonic provinces (adapted from Cordani and Teixeira, 2007; Bettencourt et al., 2010): 1—Phanerozoic sedimentary covers; 2—Precambrian sedimentary covers; 3—Central Amazonian Province (>2600 Ma); 4—Maroni-Itacaiunas Province (2250–2050 Ma); 5—Ventuari-Tapajós Province (1980–1810 Ma); 6—Rio Negro-Juruena Province (1780–1600 Ma); 7—Rondonian-San Ignacio Province (1600–1300 Ma); 8—Sunsás-Aguapeí Province (1200–950 Ma). (b) General geometry of SW Amazonian Craton showing the (ca. 1.2–1.0 Ga) Sunsás, Aguapeí and Nova Brasilândia belts (adapted from D'Agrella-Filho et al., 2008). JPSZ—Ji-Paraná Shear Zone. (c) Simplified geologic-tectonic map of the studied area—SW Mato Grosso State—(modified from Bettencourt et al., 2010). Yellow rectangles indicate two of the mafic bodies included in the Figueira Branca Intrusive Suite. The studied Indiavaí intrusion crops out in area #2, as highlighted by the rectangle (this area is shown in detail in Fig. 2).

solar orientation was normally employed due to the very high NRM intensities of the gabbro. Cylindrical specimens of 2.2×2.5 cm were cut from each sample. For selected samples an additional chip was cut from the same cylinders for optical microscopy observations.

Specimens were submitted to stepwise thermal and alternating field (AF) demagnetization in the paleomagnetic laboratories of the University of São Paulo (USP), Brazil, and the Luleå University of Technology, Sweden. In both laboratories, magnetic remanence was measured using a 2 G cryogenic magnetometer after step-wise treatment by alternating fields (AF) or thermal demagnetization. AF demagnetization was carried out using an automated three-axis AF-demagnetizer (upper limit of 170 mT) and thermal demagnetization was carried out in a Magnetic Measurements (MMTD60) or a Schonsted oven, with peak temperatures within ±2 °C and a heating time of 1 h. For samples with very high NRM intensities a Molspin Spinner magnetometer was used to measure remanent magnetization. Such samples were demagnetized using a Molspin AF-demagnetizer. These instruments are housed at the Universidade de São Paulo in a magnetically shielded room with ambient field <1000 nT. Magnetic components for each specimen were identified in orthogonal plots, and calculated using least-squares fits (Kirshvink, 1980) from a minimum of four demagnetization steps. Line-fit was filtered using an upper limit of 8° for mean angular deviation (MAD), although vectors were generally very-well defined (MAD<4°). Fisherian statistics (Fisher, 1953) were used to calculate vector mean directions and paleomagnetic poles. Additional magnetic measurements were performed in a Moslpin vibrating sample magnetometer for hysteresis curves, and a Magnetic Measurements pulse magnetizer was used for isothermal remanent magnetization

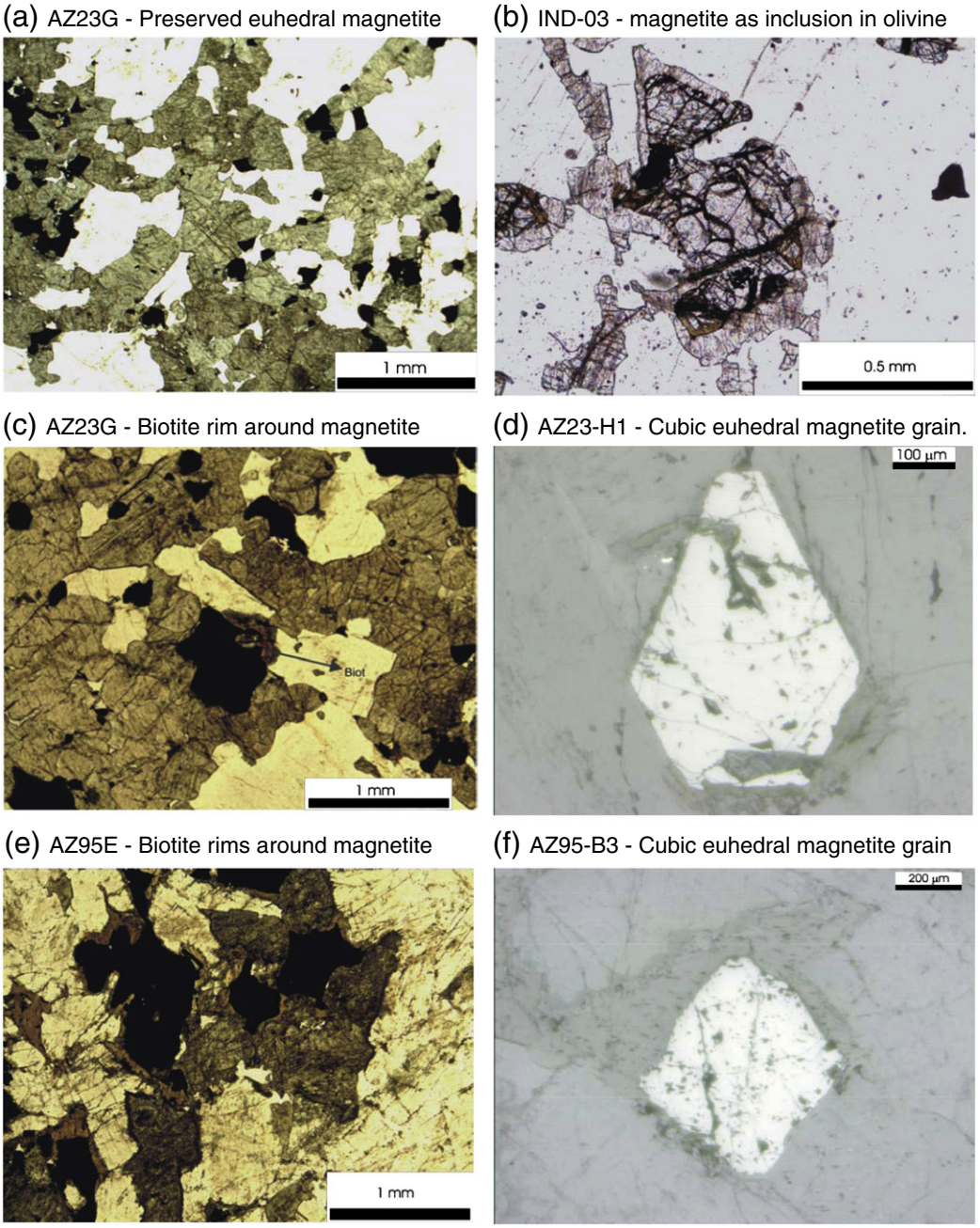


Fig. 3. Photomicrograph of gabbros from the Indiavaí Intrusion (a to d) and of a metabasic rock from the Alto Jauru Group (e and f). Panels a, b, c and e are transmitted light microscopy. d and f are reflected light microscopy. All photos are with parallel polarizers.

(IRM) acquisition curves. Low and high-temperature thermomagnetic curves were obtained for several samples using a CS-4 apparatus coupled to a KLY-4S Kappabridge instrument (Agico, Czech Republic).

4. Results

4.1. Indiavaí gabbros

4.1.1. Petrography

The mafic rocks are usually medium-grained showing intergranular texture. They essentially comprise fractured olivine surrounded by orthopyroxene and plagioclase (on average, 70% of plagioclase, 20% of olivine and 10% of pyroxene). Opaque minerals are dominated by cubic, euhedral magnetite (Fig. 3a) that occurs as inclusions in pyroxene and olivine (Fig. 3b) attesting to their primary origin during magmatic crystallization. Locally, rims of biotite are observed around magnetite grains (Fig. 3c) probably as a late-crystallizing phase in a fluid-rich magma. Preserved magnetite predominates in the rock (Fig. 3d), although very small amounts of secondary hematite due to martitization are also observed in some grains.

4.1.2. Magnetic carriers

Thermal demagnetization revealed a discrete range of unblocking temperatures (T_{ub}) for most of the sites, where more than 70% of the NRM was erased in temperatures within ~50–70 °C below the Curie temperature of magnetite (Fig. 4a). Samples from only two sites (AZ23 and AZ103) yielded a wide T_{ub} spectrum, where more than 70% of the NRM was removed at temperatures below 500 °C (Fig. 4a). Both reversible and irreversible thermomagnetic curves were found, most of them with a characteristic Verwey transition at low temperatures and a pronounced Hopkinson peak at high temperatures, just below a strong drop in susceptibility at around 580 °C, suggesting SD/PSD magnetite as the main magnetic carrier in the rocks (Fig. 5a).

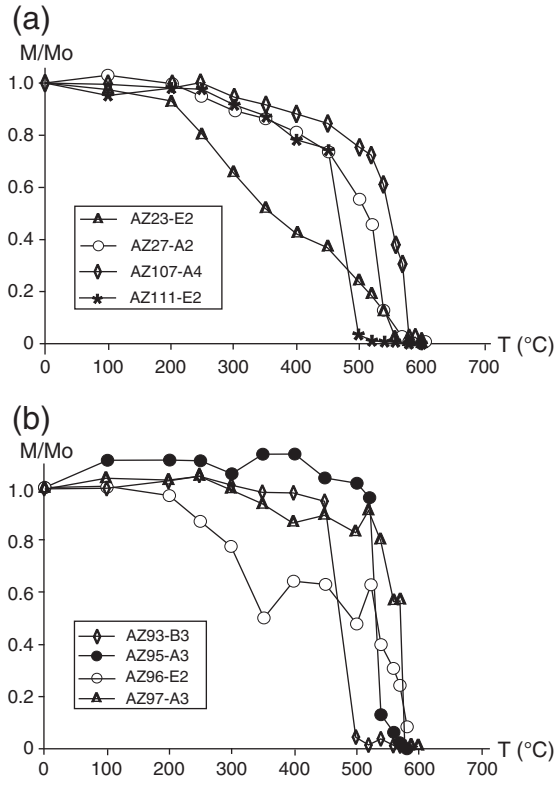


Fig. 4. Normalized intensities of natural remanent magnetization after thermal treatment for samples from the Indiavaí gabbros (a) and basement rocks (b).

Isothermal remanent magnetization (IRM) acquisition curves (Fig. 6a) show saturation fields less than 300 mT, which are also typical of magnetite. Examples of hysteresis loops performed for selected samples of the gabbros are shown in Fig. 7a, b. They give bulk coercive force (H_c), saturation magnetization (M_s), and saturation isothermal remanence (M_{rs}), after subtraction of the paramagnetic contribution from the high field portion of the curve. Coercivity of remanence (H_{cr}) was obtained using the same Molspin VSM instrument. Coercivities (H_c) vary from 5.2 to 11.5 mT, which are also typical of magnetite. In the Day's diagram (Day et al., 1977), which plots the M_{rs}/M_s and H_{cr}/H_c ratios, most gabbro samples fall along a trend parallel to the theoretical SD–MD magnetite mixing curves of Dunlop (2002a,b), with only one sample plotting in the MD field (Fig. 7e). Some samples fall in the SP–SD field, close to or between the theoretical curves traced for different percentages of 10 nm and 15 nm SP grains, respectively.

4.1.3. Paleomagnetism

The Indiavaí gabbros carry natural remanent magnetization (NRM) intensities varying from 1 A/m up to values as high as 340 A/m. Such a big variation in intensity is observed even between samples from the same site; for example, intensities vary from 1 A/m up to 134 A/m for samples from site AZ23. The high intensities observed in some samples are probably related to lightning, although this effect did not prevent the effectiveness of AF demagnetization in isolating the characteristic remanent magnetization (ChRM) in some of the samples (see example of sample AZ23G3 in Fig. 8a).

In several cases a single vector component is present in the sample (AZ28A1 in Fig. 8f), and the thermal treatment isolated a southwesterly positive high temperature component. For some sites, however, the thermal treatment does not completely remove a secondary component even close to the Curie temperature of magnetite. Overlapping of T_{ub} spectra associated to components carried by MD (low coercivities component) and SD (high coercivities component) magnetite grains (Fig. 7e) is probably the reason for this, i.e. the thermal demagnetization shows only an intermediate direction in the samples (Dunlop and Özdemir, 1997). In contrast, alternating field (AF) demagnetization was efficient in isolating the ChRM direction in the same rocks, which is normally associated with high coercivity, SD-magnetite grains. After elimination of softer directionally scattered components in fields below 30 mT, southwesterly positive inclination directions were isolated in 174 specimens from 16 sites (Fig. 8, Table 1).

Sample and site mean directions are shown in Fig. 9a. A mean ChRM direction (mean of site mean directions, Table 1) was calculated at $D_m = 209.8°$; $I_m = 50.7°$ ($N = 16$, $\alpha_{95} = 8.0°$, $K = 22$), which yielded a paleomagnetic pole (IG) at 249.7°E; 57.0°S ($A_{95} = 8.6°$, $K = 20$). Site mean directions were calculated considering each specimen direction as an independent entry since only one or two directions were frequently available for each sample (cylinder). However, the procedure of calculating sample (cylinder) mean directions before calculating the site mean directions yielded a very similar ChRM mean direction ($Dm = 208.2°$; $Im = 49.4°$; $N = 16$; $\alpha_{95} = 7.9°$, $K = 24.6$) for the Indiavaí Intrusion. The statistical parameters associated with the moderate dispersion of site mean directions suggest that secular variation has been averaged out.

4.2. Basement rocks

Unfortunately, the five sites (sites AZ21, 22, 24, 31, and 108) close to the Indiavaí Intrusion showed inconsistent within-site results, and a complete baked contact test could not be established for the Indiavaí rocks. The metabasic rocks from the Alto Jauru Group (sites AZ89 to AZ101), however, yielded consistent results for 6 out of 13 analyzed sites, for which petrography, and magnetic analysis are presented below.

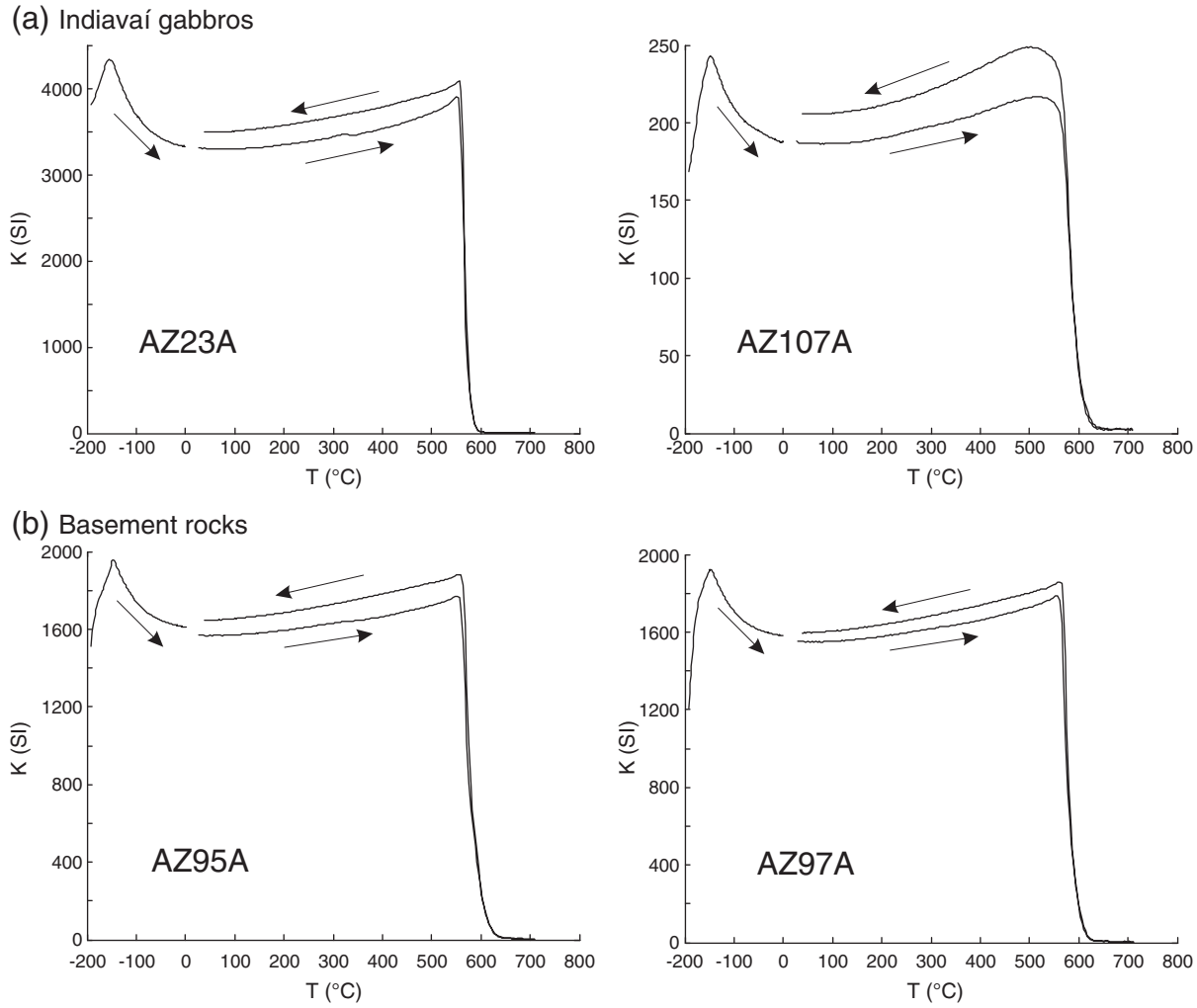


Fig. 5. Typical thermomagnetic curves showing variation in magnetic susceptibility with low and high temperature for samples from the Indiavaí gabbros (a) and basement rocks (b).

4.2.1. Petrography

The metabasic rocks show preserved sub-ophitic texture consisting of prismatic crystals of plagioclase filled by pyroxenes in their interstices. Incipient substitution of pyroxene by actinolite is observed, which indicates that these rocks were somewhat affected by metamorphism or hydrothermal alteration (Fig. 3e). These rocks contain an abundance of opaque minerals (magnetite) as inclusions in the pyroxene or as anhedral grains with rims of biotite or as pristine subhedral grains in the matrix (Fig. 3f). Some magnetite grains present fine ilmenite lamellae, diagnostic of high temperature oxyexsolution, and about 10% of the magnetite crystals present evidence of martitization forming very small amounts of hematite. Sometimes, hematite intergrowth within microfractures is also observed.

4.2.2. Magnetic carriers

Thermal demagnetization revealed a discrete range of unblocking temperatures (T_{ub}) for all analyzed samples, where more than 60% of the NRM was erased in temperatures within ~50–70 °C below Curie temperatures typical of magnetite (Fig. 4b). Both reversible and irreversible thermomagnetic curves have been found, most of them with a pronounced Hopkinson peak, a characteristic Verwey transition, and Curie temperatures around 580 °C, suggesting SD/PSD magnetite as the main magnetic carrier (Fig. 5b). However, for some of the samples the decay of remanence during thermal demagnetization starts at T_{ub} between 200 °C and 350 °C (Fig. 4b) suggesting the presence of iron sulfides (pyrrhotite?) in these rocks besides magnetite. Isothermal remanent magnetization (IRM) acquisition curves (Fig. 6b) show saturation fields less than 300 mT which are also typical for magnetite minerals. Examples of hysteresis loops performed on selected samples are shown in Fig. 7c, and d. The coercivities are generally low and vary from 1.8 to 14.7 mT. In the Day's diagram (Fig. 7e) most samples fall along a trend parallel to the theoretical SD–MD magnetite mixing curves (Dunlop, 2002a,b) with two samples plotting in the MD field. Some samples plot close to the SP (10 nm)–SD mixing curve defined by Dunlop (2002a).

4.2.3. Paleomagnetism

The basement rocks present natural remanent magnetization (NRM) intensities varying from 4 A/m up to 57 A/m. Except for single vector component samples (e.g., sample AZ93B3 in Fig. 10), the thermal treatment did not completely remove secondary components even close to the Curie temperature of titanomagnetite. AF demagnetization was more efficient to isolate either south-southwestern, upward inclination directions or northwestern, downward inclination directions after removal of low-coercivity secondary components (Fig. 10, Table 2).

Sample and site mean directions are shown in Fig. 9b. Consistent within site directions were isolated for samples from only six sites (Table 2), and among these the downward site mean directions are more scattered than the upward directions (Fig. 9b). Also, these directions do not form exactly antipodal groups of reversed and normal

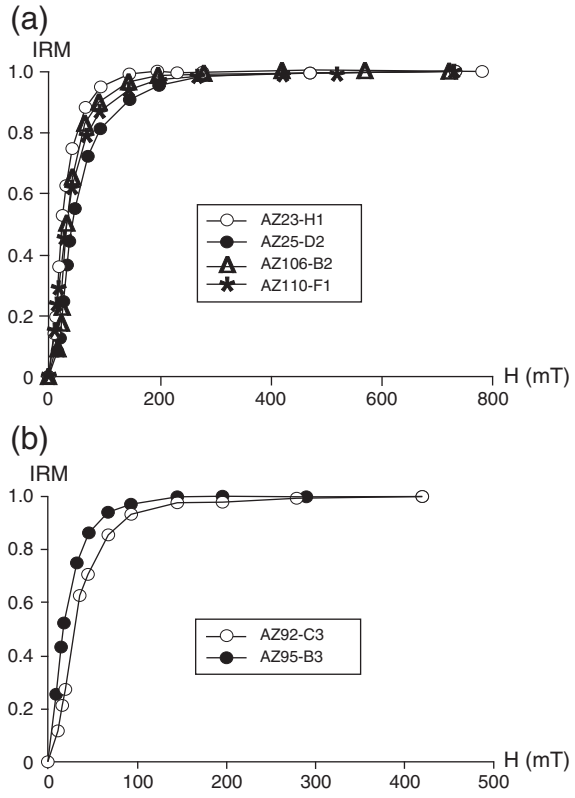


Fig. 6. IRM acquisition curves for samples from the Indiavaí gabbros (a) and basement rocks (b).

directions suggesting they recorded the geomagnetic field at quite different times. The downward directions seem to define two groups of directions, which are denoted as components A and B, and the upward directions are denoted as component C. Although the basement rocks yielded more complicated results than the Indiavaí gabbros, the A, B and C components are significantly different from both the present geomagnetic field and from the southwestern, downward directions found in the intrusions, located ca. 10 km from the sites of the basement rocks. This demonstrates that no regional event of remagnetization affected the studied area after the Indiavaí magmatic event.

5. Discussion

5.1. Paleomagnetic pole of the Indiavaí gabbros

'Reversed' south/southwestern stable remanence directions were disclosed in samples from the Indiavaí mafic intrusions. The following evidences suggest a primary origin for this remanence component: (i) the magnetic experiments indicate it is carried by SD/PSD magnetite grains with moderate to high H_c and high-unblocking temperatures (>500 °C), denoting its high magnetic stability; (ii) the petrography shows magmatic cubic euhedral magnetites as the main magnetic mineral; rims of biotite around the magnetite probably formed at the late crystallization stages are the only observed secondary mineral paragenesis in the gabbros; (iii) country rocks located ca. 10 km to NW of the Indiavaí mafic intrusions carry remanent magnetization with different directions (Fig. 9), which demonstrate that probably no later event of remagnetization has occurred in the region after cratonization. Note that these directions are also significantly different from the present geomagnetic field, but they are similar to remanence directions isolated in roughly coeval rocks from other parts of Amazonia: mean directions Dec. = 334.0°, Inc. = 27.0° (N = 7, α_{95} = 11°) and Dec. = 325.5°, Inc. = 23.5° (N = 11, α_{95} = 11°) were obtained, respectively, for diabases from Guyana and Venezuela (Hargraves, 1968), and from sites of the Rio Aro (2) and Rio Guaniamo (1) dykes, and from La Escalera (8) diabases in Venezuela (Onstott et al., 1984). These authors describe these directions as Component I for which they attributed an age of 1640 Ma, based on Rb–Sr geochronology. Also, Veldkamp et al. (1971) published a mean direction determined for 30 samples (two sites) of dolerites from Guyana and Venezuela (Dec. = 314.0°, Inc. = 3.5°, α_{95} = 3°). An age of ca. 1750 Ma was attributed for this magnetic component by the authors. These facts imply that the characteristic paleomagnetic component of the Indiavaí rocks is most probably a thermo-remanent magnetization acquired shortly after the intrusion at about 1415.9 ± 6.9 Ma ago, as determined by the U–Pb SHRIMP analysis (Teixeira et al., 2011). Additional, independent evidence for the primary nature of the remanence isolated in the gabbro is the fact that similar remanence directions were obtained for the Nova Guarita mafic dykes that crop out ca. 600 km NE of the Indiavaí region, for which the ^{40}Ar–^{39}Ar age of 1418.5 ± 3.5 Ma overlaps the age of the Indiavaí intrusion (Bispo-Santos et al., 2012). This fact also demonstrates that no local tilting effects occurred between these distant areas.

The paleomagnetic pole of the Indiavaí gabbros (IG pole: 247.7°E, 57.0°S, N = 16, A_{95} = 8.6°, K = 19.6) satisfies 4 out of the 7 quality criteria proposed by Van der Voo (1990) as follows: (1) The U–Pb (SHRIMP) zircon crystallization age of 1415.9 ± 6.9 Ma probably represents the age of the primary thermoremanent magnetization acquired by the analyzed gabbros (see discussion below); (2) the IG pole was derived from a set of 174 specimens (16 sites) and has adequate Fisher's statistical parameters (A_{95} = 8.6°, K = 20); (3) remanence vectors were isolated by least-squares fit from orthogonal diagrams (Kirshvink, 1980) after stepwise AF demagnetization; (5) the Indiavaí Intrusion (and coeval intrusive bodies of the Figueira Branca Intrusive Suite) is contemporaneous with the late orogenic phase of the Santa Helena arc (1460–1420 Ma)—one of the accretionary arcs tectonically related to the Rondonian-San Ignacio Province (Bettencourt et al., 2010). It represents the magmatism linked with an extensional regime over the tectonically stable Jauru Terrane (Orosirian in age). After this magmatism the area became stable, subjected only to uplift, cooling processes, and finally to tectonic reactivations along the Aguapeí aulacogen that occurs to the west of the Indiavaí intrusion (Teixeira et al., 2011). Moreover, the consistency between IG and NG poles also supports the idea that neither result is significantly affected by local tilting.

Nevertheless, three criteria are not wholly satisfied: (4) a positive result for the baked contact test could not be demonstrated, although the different directions found for the basement rocks, probably excludes a later regional remagnetization effect; (6) only one polarity was recorded; and (7) similar directions were found for 980 Ma mafic sills from the Rio Branco area and 1150 Ma sediments from the Aguapeí Group (D'Agrella-Filho et al., 2008; Elming et al., 2009b). Although the Rio Branco area is located approximately 55 km to the east of the studied area, the different host rock magnetization in our study (see above) implies it is very unlikely that a remagnetization of the Rio Branco age could have remagnetized the Indiavaí gabbros and left the magnetization of the country rocks preserved. Moreover, a similar direction was also found for the 1420 Ma Nova Guarita dykes (Bispo-Santos et al., 2012), for which a positive baked contact test assures its primary origin. The Indiavaí pole (IG) may thus be qualified with Q = 4 and it can be used to constrain the paleogeographic position of the proto-Amazonian Craton in the Columbia supercontinent at ~1416 Ma.

5.2. Age of remanence acquisition in the Indiavaí gabbros

Although several lines of evidence suggest a primary origin for the IG paleomagnetic pole with age at 1415.9 ± 6.9 Ma, biotites of two samples from Indiavaí gabbros dated by the ^{40}Ar–^{39}Ar method

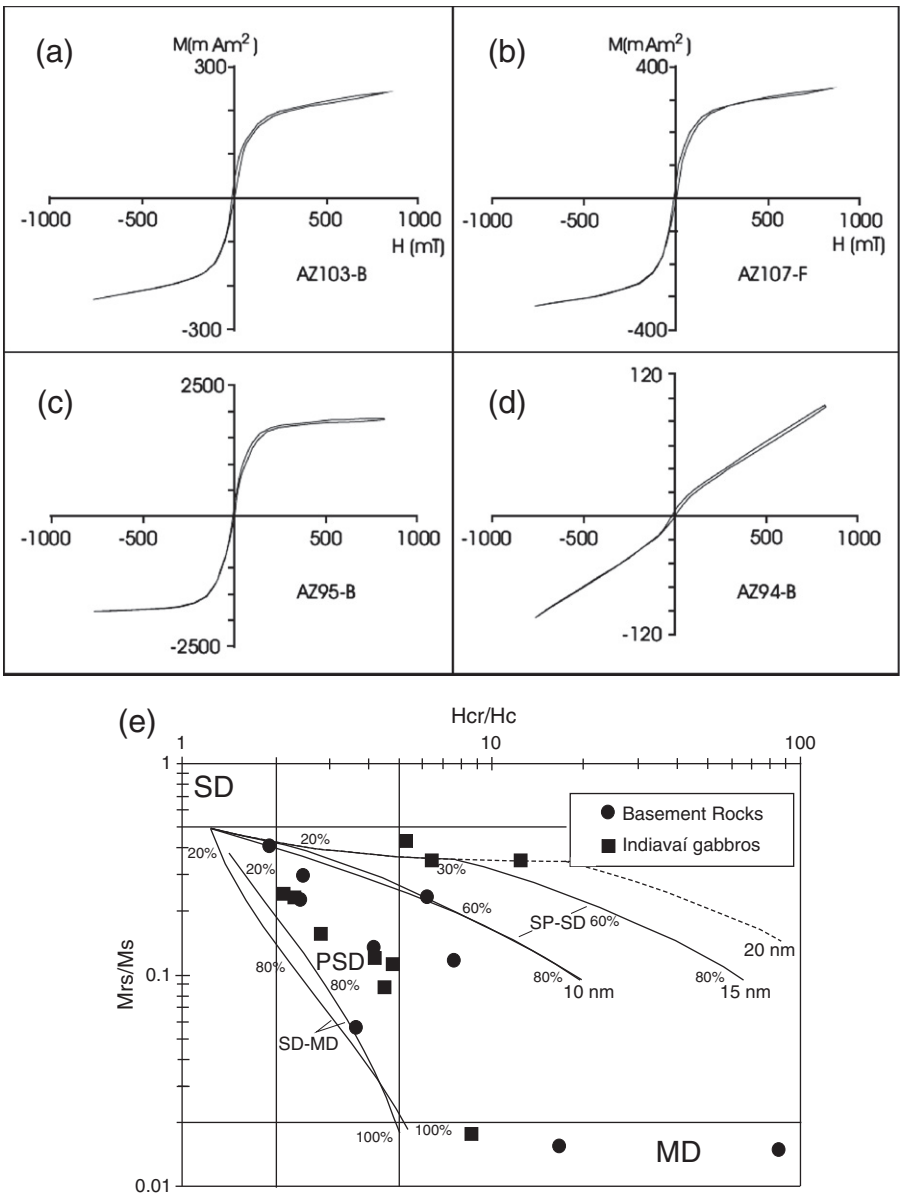


Fig. 7. Typical hysteresis loops obtained for samples from the Indiavaí gabbros (a, b), and for samples from basement rocks (c, d). (e) Day's diagram: M_{rs}/M_r versus H_{cr}/H_c ratios (Day et al., 1977) obtained for samples from both Indiavaí gabbros and basement rocks. The limits between SD, PSD and MD fields are those proposed by Dunlop (2002a,b). SD + MD and SP + SD (with SP grain sizes of 10, 15, and 20 nm) mixing curves are also shown. Most samples are compatible with SD plus MD mixing curves as proposed by Dunlop (2002a,b).

(AZ23 and AZ110, see Table 1) yielded much younger plateau ages of 1222 ± 3 Ma and 1222 ± 4 Ma for a sample from site AZ23, and apparent ages of 1275 ± 4 Ma and 1268 ± 4 Ma for a sample from site AZ110 (Teixeira et al., 2011). This indicates that the biotite system was closed for Ar diffusion ca. 140–200 Ma after rock crystallization. The simplest interpretation should be that a local thermo-tectonic event could be responsible for a temperature increase up to biotite argon-unblocking temperatures (300–350 °C, McDougall and Harrison, 1999) at that time. Reactivation of large, crustal-scale structures reflects events that may have disturbed the ^{40}Ar–^{39}Ar system in the region. To the northwest of the studied area, the Ji-Paraná strike-slip shear zone (Fig. 1b) has been active between 1200 Ma and 1150 Ma affecting the Amazonian Craton basement to the north of the Nova Brasilândia belt (Fig. 1) (Tohver et al., 2005b, 2006). On the other hand, ^{40}Ar–^{39}Ar ages obtained for rocks to the west of the studied area are much younger than those obtained for the Indiavaí gabbros, and fall between 1000 and 900 Ma (Geraldes et al., 2001; Ruiz, 2005; Tohver et al., 2006). They are associated with the main phase of deformation of the Aguapeí Group (Teixeira et al., 2010). Unfortunately, we have no U–Pb or ^{40}Ar–^{39}Ar ages on other minerals (such as titanite or hornblende) in the investigated rocks to better define the cooling history of the Indiavaí Intrusion. Based on the regional temperature–time paths defined for the Jauru polycyclic domain (Teixeira et al., 2011), short cooling histories for the Cachoeirinha (1590–1520 Ma) and

Fig. 8. Examples of AF and thermal demagnetization for samples from the Indiavaí gabbros. Stereographic projections (full (empty) symbols represent downward (upward) inclinations); orthogonal projections, and magnetization intensity decay curves ($M/M_o \times H$). The respective H_c or T_{ub} spectra are also shown.

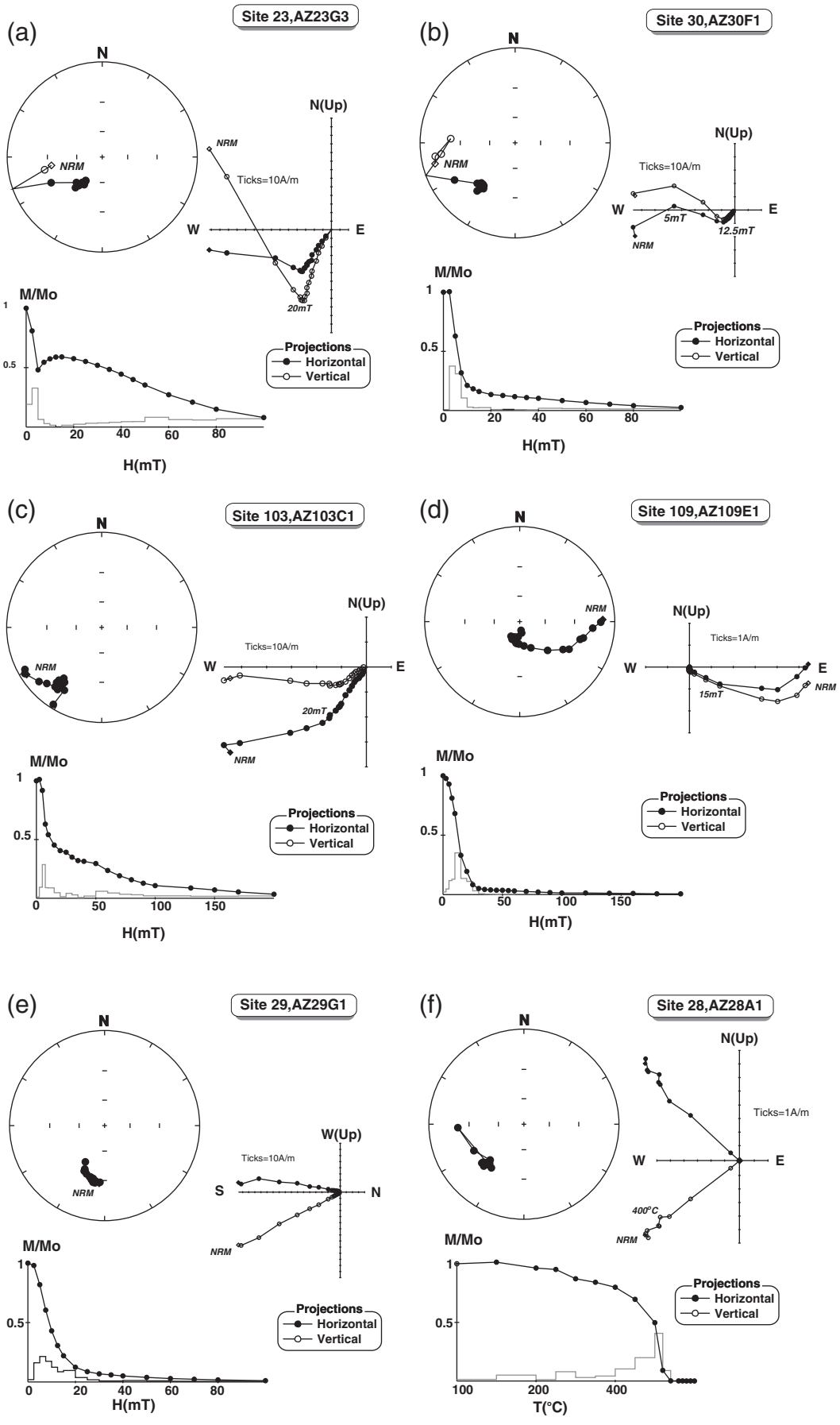

(a)
Site 23,AZ23G3
N
NRM
N(Up)
NRM
Ticks=10A/m
W
E
20mT
M/Mo
1
0.5
0
20
40
60
80
H(mT)
Projections
Horizontal
Vertical
(b)
Site 30,AZ30F1
N
NRM
N(Up)
Ticks=10A/m
W
E
5mT
12.5mT
NRM
M/Mo
1
0.5
0
20
40
60
80
H(mT)
Projections
Horizontal
Vertical
(c)
Site 103,AZ103C1
N
NRM
N(Up)
Ticks=10A/m
W
E
20mT
NRM
M/Mo
1
0.5
0
50
100
150
H(mT)
Projections
Horizontal
Vertical
(d)
Site 109,AZ109E1
N
NRM
N(Up)
Ticks=1A/m
W
E
15mT
NRM
M/Mo
1
0.5
0
50
100
150
H(mT)
Projections
Horizontal
Vertical
(e)
Site 29,AZ29G1
N
NRM
W(Up)
Ticks=10A/m
S
N
NRM
M/Mo
1
0.5
0
20
40
60
80
H(mT)
Projections
Horizontal
Vertical
(f)
Site 28,AZ28A1
N
N(Up)
Ticks=1A/m
W
E
400°C
NRM
M/Mo
1
0.5
100
200
400
T(°C)
Projections
Horizontal
Vertical

Table 1
Paleomagnetic results for the Indiavaí Intrusive rocks.

				Mean site direction					VGP	
Site	N	Lat (°S)	Long (°W)	n/n_u	Dec°	Inc°	k	α_{95}°	Plong (°E)	Plat (°N)
AZ-23	8	15.457	58.585	15/14	212.1	58.8	143.0	3.3	258.4	−53.1
AZ-25	8	15.465	58.581	16/12	200.2	40.9	26.1	8.7	237.2	−69.4
AZ-26	8	15.465	58.581	14/11	198.3	58.4	33.0	8.1	270.8	−61.5
AZ-27	9	15.465	58.581	18/08	262.4	61.2	27.7	10.7	251.7	−15.8
AZ-28	8	15.465	58.581	19/11	200.5	29.4	37.0	7.6	215.0	−70.3
AZ-29	7	15.465	58.581	17/15	214.4	38.0	10.4	14.5	226.9	−56.9
AZ-30	6	15.468	58.582	15/11	197.9	32.1	25.2	9.3	220.5	−72.7
AZ-102	6	15.455	58.585	10/07	206.0	59.8	48.7	8.7	264.7	−56.2
AZ-103	7	15.455	58.584	16/12	209.5	56.4	40.2	6.9	256.5	−56.2
AZ-104	6	15.457	58.583	14/13	209.2	47.7	21.1	9.2	242.7	−60.0
AZ-105	6	15.460	58.582	17/12	210.8	18.0	57.4	5.8	203.4	−59.3
AZ-106	6	15.461	58.581	13/12	204.1	52.3	22.9	9.3	254.5	−62.1
AZ-107	6	15.462	58.578	11/09	202.9	50.0	32.7	9.1	251.7	−64.1
AZ-109	6	15.469	58.578	11/09	207.8	67.3	180.9	3.8	274.5	−48.7
AZ-110	6	15.467	58.578	11/09	213.0	54.9	37.3	8.5	251.8	−54.4
AZ-111	7	15.466	58.575	11/09	248.8	69.1	100.2	5.2	262.8	−25.1
Mean				16	209.8	50.7	22.1	8.0	249.7	−57.0
									K = 19.6	α_{95} = 8.6°

N (number of collected samples); Lat/Long (geographic coordinates); n/n_u (number of analyzed specimens/number of specimens used in mean directions); Dec (declination); Inc (inclination); α_{95} (Fisher's cone of confidence); K (Fisher's precision parameter); VGP (virtual geomagnetic pole); Plong (pole longitude); Plat (pole latitude).

Santa Helena (1490–1420 Ma) orogens can be envisaged. This suggests a fast cooling of the Indiavaí intrusive host rocks, which is also coherent with our petrographic analysis that shows no evidence of metamorphism in these medium grain sized rocks. On the other hand, if the ^{40}Ar–^{39}Ar ages for the Indiavaí gabbros resulted from a slow cooling following cratonization of the area at about 1300–1250 Ma, it must be noticed that the narrow and elevated blocking temperature of magnetic carriers found in most sites would record the magnetic field direction at high temperatures. The time dependence of blocking temperature (time versus blocking temperature curves of Pullaiah et al., 1975) shows that even for the extreme case of 200 Ma (2×10^8 years) burial at 300 °C–350 °C, the 1 h laboratory blocking temperatures above 450 °C recorded by the Indiavaí gabbros are still likely to preserve their primary remanence. Unblocking temperatures between 500 °C and 580 °C are associated with almost all studied sites. From the above, the characteristic remanent magnetization (IG component) isolated after AF treatment for the Indiavaí samples is interpreted as a primary thermo-remanent magnetization acquired at about 1416 Ma ago.

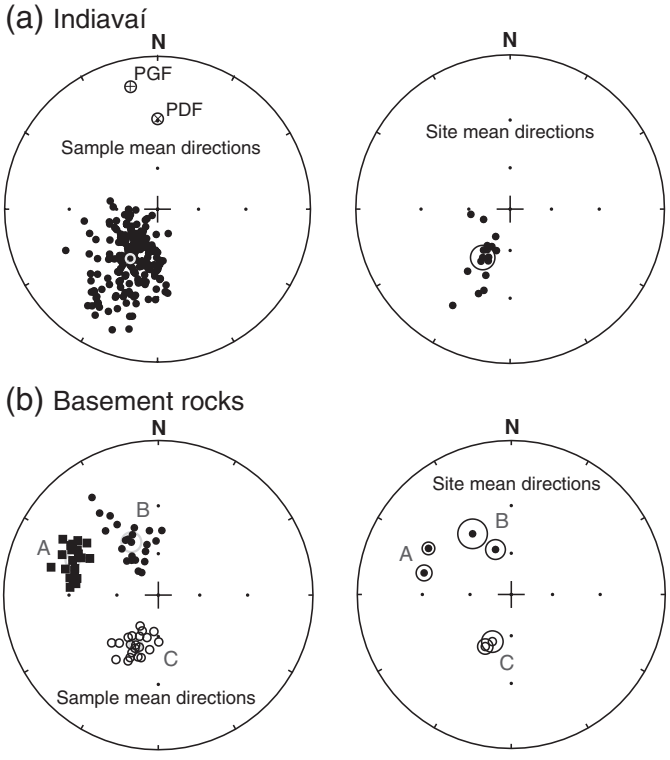


Fig. 9. Sample and site mean directions for the Indiavaí gabbros (a) and basement rocks (b). Full (empty) symbols represent downward (upward) inclinations. PDF—Present Dipolar Field; PGF—Present Geomagnetic Field. A, B, and C are basement components as defined in the text. Confidence circles around the means are also shown.

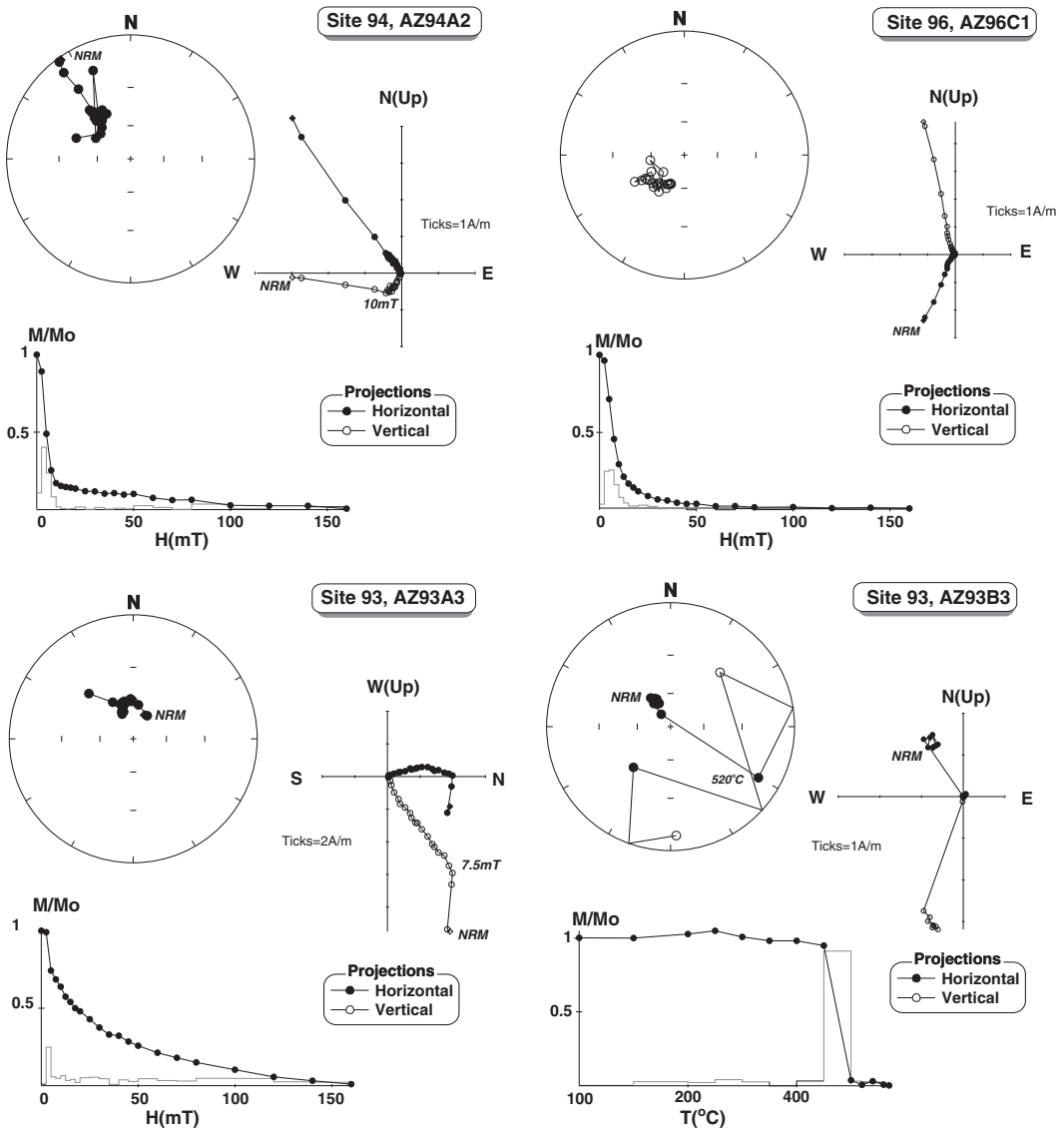


Fig. 10. Examples of AF and thermal demagnetization for samples from the basement rocks. Stereographic projections (full (empty) symbols represent downward (upward) inclinations); orthogonal projections, and magnetization intensity decay curves ($M/M_o \times H$). The respective H_c or T_{ub} spectra are also shown.

5.3. Significance of the Nova Brasilândia belt (NBB)

Based on geophysical data and structural inferences Tohver et al. (2004) interpreted the 1100–1000 Ma Nova Brasilândia metasedimentary belt (NBB; Fig. 1b) to be an E–W suture zone originating from the collision of the Paraguá Terrane and the proto-Amazonian Craton. The basis for this conclusion is the strong contrast between the monocyclic history of the NBB, composed of high-pressure–temperature (P–T) granulitic assemblages (1090–1060 Ma), and the polycyclic Amazonian basement to the north, with structural and geochronological evidence for sinistral strike-slip deformation at 1190–1130 Ma (Ji-Paraná shear zone). Noteworth, deformation of this age is not recognized in the younger, 1100–1000 Ma NBB. The NBB itself comprises part of a large-scale E–W trending magnetic lineament, approximately 1000 km long, that continues beneath the Serra dos Parecis sedimentary rocks to the east (Fig. 1c). Tohver et al. (2004) postulated that the Jauru Terrane, where the Indiavaí Intrusion crops out, is part of the Paraguá Terrane, which may have been separated from the Amazonian Craton until the two were sutured across the NBB at ca. 1000 Ma ago. This interpretation contrasts with the suggestion that the Jauru Terrane is an extension to the southeast of the Rondonian-San Ignacio (RSI) Province (see previous section), and also, that the Paraguá Terrane (as defined by Litherland et al., 1986) collided with the proto-Amazonian Craton along the Rio Alegre Terrane at about 1350–1300 Ma ago (Bettencourt et al., 2010).

Since the two recently studied mafic intrusions (Nova Guarita and Indiavaí) are coeval and located on different sides of the NBB, the Nova Guarita dyke swarm to the north (at latitudes around 10°S; Bispo-Santos et al., 2012) and the Indiavaí gabbros (at latitudes around 15.5°S) to the south of the proposed NBB suture (located at latitudes between 11°S and 12°S), they can be used to test this hypothesis. Fig. 11 shows these two poles for the Amazonian Craton in its present position. Since the poles overlap within error, no large scale latitudinal tectonic difference or relative rotations are indicated, which may suggest that the NBB likely resulted from internal tectonic reactivations during the collision of Amazonia with Laurentia along the Sunsás-Grenville orogeny (e.g., Cordani et al., 2009). One may argue that paleomagnetism does not resolve paleolongitudinal movements. Indeed, an overlap of these poles is also possible if there was a nearly E–W approaching movement (since ~1420 Ma ago) between the Paraguá Terrane and the proto-Amazonian Craton, a movement that may be at the origin of the 1100–1000 Ma NBB. However, the Nova Guarita and Indiavaí poles in Fig. 11, show that the NBB strikes approximately parallel to a line of paleolatitude. Thus, the

Table 2
Paleomagnetic results for the basement rocks.

Site	N	Lat (°S)	Long (°W)	Site mean direction					VGP	
				n/n_u	Dec°	Inc°	k	α_{95}°	Plong (°E)	Plat (°N)
AZ-92(C)	6	15.375	58.652	12/10	201.9	−53.5	46.1	7.2	143.9	−36.4
AZ-93 (B)	6	15.376	58.652	13/11	340.7	55.6	48.0	6.7	282.3	35.3
AZ-94 (B)	6	15.375	58.652	13/11	327.5	39.9	26.7	9.0	260.7	40.4
AZ-95 (A)	6	15.375	58.650	12/08	299.1	26.8	182.0	3.4	234.5	23.0
AZ-96 (C)	6	15.374	58.649	10/10	206.5	−48.6	95.5	5.0	151.0	−38.3
AZ-97 (A)	7	15.374	58.649	11/10	283.9	29.5	123.2	4.4	230.5	8.7

N (number of collected samples); Lat/Long (geographic coordinates); n/n_u (number of analyzed specimens/number of specimens used in mean directions); Dec (declination); Inc (inclination); α_{95} (Fisher's cone of confidence); K (Fisher's precision parameter); VGP (virtual geomagnetic pole); Plong (pole longitude); Plat (pole latitude). Isolated components (A), (B), and (C) are indicated for each site (first column) as described in the text and Fig. 9b.

overlapping poles at 1420 Ma suggest that the Indiavaí and Nova Guarita areas were not separated by any significant paleolatitudinal distance at that time.

This last observation is a key constraint on the significance of the NBB in terms of defining whether the Amazonian Craton was not continuous with the Paraguá Craton in Paleo-Mesoproterozoic times. At stake is the controversy over whether the Paleoproterozoic to Mesoproterozoic geochronological provinces of the Amazonian Craton are coherent across the NBB, except for offsets generated during late Mesoproterozoic activity. Two possibilities are envisaged: first, that the NBB was created by strike-slip emplacement with limited paleolatitudinal convergence between previously separate blocks, or second, that the NBB is a failed branch of one continental rift in an otherwise conjoined Amazon-Paraguá Craton. Following the discussion of Tohver et al. (2004), who regarded the NBB as a suture zone between two, previously-separate blocks, the ages of deposition for both the Aguapeí and Nova Brasilândia original basins are broadly similar, with a maximum age established by the youngest detrital zircon by means of U/Pb geochronology. A youngest zircon age of 1215 ± 20 Ma was reported by Santos et al. (2000) for the NBB, whereas Santos et al. (2001) and Leite and Saes (2003) reported a youngest detrital zircon age of 1165 ± 27 Ma for the basal formations of the Aguapeí Group. Subsequently, D'Agrella-Filho et al. (2008) reported a diagenetic U–Pb age of 1149 ± 7 Ma from xenotime overgrowths on detrital zircons in the Fortuna Formation at the base of the Aguapeí Group. Given the broad compatibility between the <1210 Ma age of deposition for the NBB, and the inferred 1150 Ma age of deposition for the Aguapeí Group, correlating these two basins would imply a common episode of deposition during the late Mesoproterozoic, which would have taken place in a series of intracontinental basins over the RSI crust. If so, the NBB may represent an active foreland basin or a failed rift (as has already been suggested for the Aguapeí belt), which opened sometime after the tectonic stability of the ultimate event that formed the RSI Province (ca. 1320 Ma; Bettencourt et al., 2010) and was closed at ca. 1100 Ma. We note that there is a strong, sinistral strike-slip motion recorded in the Nova Brasilândia belt (Tohver et al., 2004), but the hypothesis of a separate origin for the Amazonian and Paraguá cratons requires that this motion accommodates almost all of the convergence between these two blocks, if they were independent. This, in turn, demands minimal offset of geochronological provinces between the basement rocks on either side of the NBB.

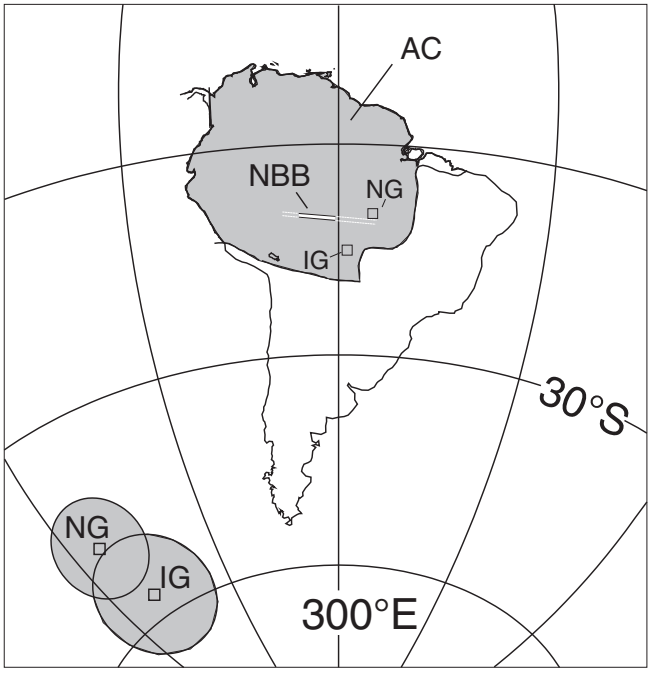


Fig. 11. Comparison of the ca. 1420 Ma Nova Guarita (NG) and Indiavaí Gabbros (IG) poles. AC—Amazonian Craton in its present position: NG—Nova Guarita area, IG—Indiavaí Gabbros area, NBB—Nova Brasilândia Belt strike. South America outline is also shown.

5.4. The Columbia supercontinent

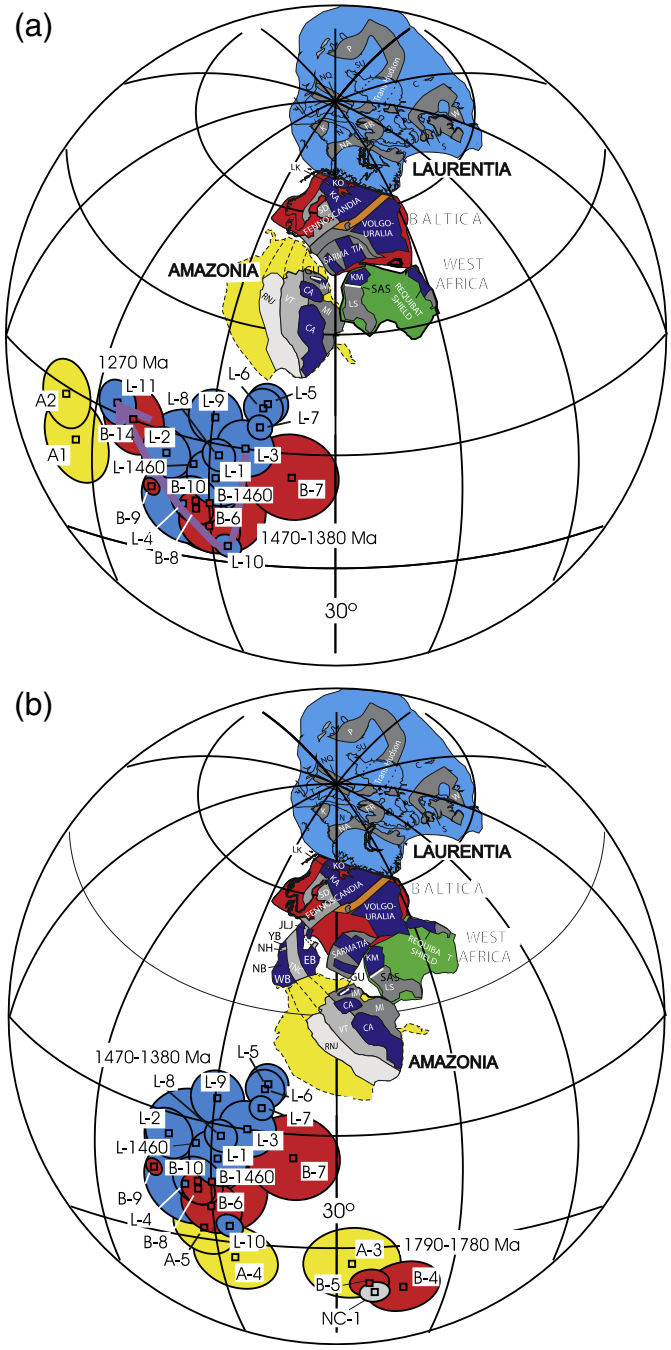
In most reconstructions of Columbia, the north-northeastern Amazonian Craton is linked to southwestern Baltica (e.g., Geraldes et al., 2001; Zhao et al., 2002, 2004, 2006; Hou et al., 2008; Johansson, 2009), and eastern Greenland is linked to northern Baltica (Buchan et al., 2000; Karlstrom et al., 2001; Pesonen et al., 2003; Zhao et al., 2006; Bispo-Santos et al., 2008; Evans and Pisarevsky, 2008; Salminen and Pesonen, 2007; Pisarevsky and Bylund, 2010; Lubnina et al., 2010). These reconstructions juxtapose age-similar Archean to Mesoproterozoic geological provinces. Based on paleomagnetic data, Buchan et al. (2000), Salminen and Pesonen (2007), Evans and Pisarevsky (2008), Pisarevsky and Bylund (2010), and Lubnina et al. (2010) suggested that Baltica/Laurentia drifted together during most of the Paleo/Mesoproterozoic (1.8 Ga up to 1.27 Ga). So, the pole of the Indiavaí Intrusive rocks (IG) represents a great opportunity to test configurations with Amazonia linked to Baltica and Laurentia.

Fig. 12a shows one possible reconstruction of Columbia, with Baltica in its present position. The Laurentia/Baltica connection (rotation pole for Laurentia: 42.9°N, 3.2°E, −48.7°) is similar to that proposed by Salminen and Pesonen (2007), and the Baltica/Amazonia link is similar to that proposed in the SAMBA model (Johansson, 2009). In this reconstruction, West Africa is shown rotated to Baltica according to the Euler pole (06°N, 029°E, −93°) proposed by Evans and Mitchell (2011), who envisage a similar model for the Baltica–Amazonia–West Africa link. Here, a slightly different Euler pole (43°N, 197°E, 84°) from that of Evans and Mitchell (52.4°N, 207.1°E, 82.8°) is used for the Amazonian Craton, since the use of the Evans and Mitchell Euler pole results in a slight overlap of Amazonia and Baltica. In our reconstruction, similar Archean and Paleo- to Mesoproterozoic geological provinces fit adequately on both cratonic blocks. In such reconstruction, the Guri and Sassandra shear zones from the Amazonian and West African cratons, respectively, are aligned, as suggested by Onstott and

Hargraves (1981) and Nomade et al. (2003), based on Paleoproterozoic paleomagnetic poles from Guyana and West African shields.

Selected paleomagnetic poles (Table 3) for the three continental blocks (Laurentia, Baltica and Amazonia) between 1470 Ma and 1270 Ma are also shown in Fig. 12a after rotating them according to rotation poles cited above. The APW path between 1470 Ma and 1270 Ma for Laurentia, proposed by Evans and Mitchell (2011) is also shown, although the presently available 1470–1400 Ma poles from Laurentia suggest a more complex APW path for this time interval (see Elston et al., 2002).

Confidence circles of the mean 1460 Ma poles for Baltica (B-1460) and Laurentia (L-1460) overlap with each other in this reconstruction, which means there is paleomagnetic support for an interpretation of these cratons being linked together at that time (see Salminen and Pesonen, 2007; Lubnina et al., 2010; Pisarevsky and Bylund, 2010; Evans and Mitchell, 2011). Confidence circles for the 1416 Ma Indiavaí Intrusion pole (AC-1416: A1, Fig. 12a) and the 1418 Ma Nova Guarita pole (AC-1418: A2, Bispo-Santos et al., 2012) from the Amazonian Craton also overlap, however, they are located ca. 30° west of the 1460 Ma mean poles representing Laurentia and Baltica. Unfortunately, there is no reliable 1420 Ma paleomagnetic pole for Baltica to provide a more definitive test. For Laurentia, some poles between 1440 Ma and 1380 Ma (L5 to L10 in Table 3) are also shown, all of them located closer to the 1460 Ma mean pole for Laurentia. Interestingly, the Indiavaí and Nova Guarita poles close to the ca. 1270 Ma poles for Laurentia and Baltica (Table 3, Fig. 12a). An age of ca. 1270 Ma for the Indiavaí pole is unlikely, even if similar ages were obtained in biotites from the Indiavaí gabbros by ^{40}Ar–^{39}Ar geochronology (see above). The same is true for the Nova Guarita dykes, which presented a positive baked contact test, and consistent, well-defined ^{40}Ar–^{39}Ar plateau ages for biotites from four studied dykes yielding a mean age of 1418.5 ± 3.5 Ma (Bispo-Santos et al., 2012). So, the Indiavaí and Nova Guarita poles do not support the Mesoproterozoic SAMBA reconstruction proposed by Johansson (2009).



5.4.1. A new scenario for the Mesoproterozoic Columbia supercontinent

Recently, Elming et al. (2010) compared Paleoproterozoic (2600–1770 Ma) poles from Fennoscandia and Ukrainian (including Sarmatia) blocks, and suggested that Baltica was not yet formed at about 1760 Ma ago. They also suggested that Sarmatia before its final docking was oriented 43° counterclockwise relative to Fennoscandia/Volgo-Uralia block, about an Euler pole at 48°N, 42.2°E, as in Fig. 12b. When rotating Amazonian and West Africa from their positions in the SAMBA configuration, using the Sarmatia rotation pole, the Indiavaí and Nova Guarita poles are more consistent with the 1470–1380 Ma Baltica and Laurentia poles. In this new configuration the 1790–1780 Ma reference poles for Amazonia (Colider pole) and Baltica (and Sarmatia) also overlap. Interestingly, inclusion of North China (Euler pole at 54.2°N, 238.2°E, −72.2°) between Baltica and Amazonian Craton (Bispo-Santos et al., 2008) is paleomagnetically allowed, when comparing the 1790–1780 Ma poles from Baltica, Amazonian Craton and North China (Fig. 12b) after rotating them according to this new configuration (see Table 3 for paleomagnetic and rotation poles). In this new scenario Archean and Paleoproterozoic geological provinces are aligned, and the same is for the Guri and Sassandra shear zones.

Kusky et al. (2007) also suggested that North China was located between Baltica and Amazonia at 1850–1800 Ma ago, and that it broke up from Columbia supercontinent soon after, at about 1780 Ma, a breakup based on profuse NW trending dyke swarms observed in

Fig. 12. (a) Columbia reconstruction at ~1460 Ma, and selected paleomagnetic poles with their corresponding cone of confidence circles (Table 3) for Baltica (blue), Laurentia (red) and Amazonia (yellow) for the time interval 1470 and 1270 Ma. Baltica is in its present position. The following Euler rotation poles were used for the other cratons and their respective poles: Laurentia—42.86°N, 3.2°E (−48.66°); Amazonian Craton—52.4°N, 207.1°E (82.8°); West Africa—6°N, 29°E, −93° (Evans and Mitchell, 2011). (b) Alternative reconstruction of Columbia, and selected paleomagnetic poles with their corresponding cone of confidence circles (Table 3) for Baltica (blue), Laurentia (red), and Amazonia (yellow) for the time interval 1470 and 1270 Ma. 1790–1780 Ma poles for proto-Baltica (and Sarmatia—see text for details), Amazonian Craton and North China are also shown (Table 3) for comparison. Proto-Baltica, Laurentia, and their respective poles, as in panel a. The following Euler rotation poles were used for the other cratons and their respective poles: Sarmatia—48.0°N, 42.2°E, 43.0° (Elming et al., 2010); Amazonian Craton—70.9°N, 255.6°E, 98.8°; West African Craton—16.5°N, 224.3°E, 67.1; North China—54.2°N, 238.2°E, −72.2°. See text for details. Archean cratonic areas: Laurentia (S—Slave; C—Churchill; SU—Superior; N—Nain and equivalent in southern Greenland), Baltica (KO—Kola; KA—Karelia), Amazonia (CA—Central Amazonia Province; IM—Imataca Complex), West Africa (KM—Kenema-Man domain), North China (YB—Yinshan block; OB—Ordos Block; WB—West Block; EB—East Block). Paleoproterozoic belts: Laurentia (NQ—New Quebec; T—Torngret; W—Wopmay; P—Penokean; K—Kefilidian; NA—Nagssugtoqidian; FR—Foxe-Rinklan), Baltica (LK—Lapland-Kola; SD—Svecofennian Domain), North China (NH—North Hebei/Khondolite belt; TNC—Trans-North China belt; JLJ—Jiao-Liao-Ji belt), Amazonia (MI—Maroni-Itacaiunas; VT—Ventuari-Tapajós; RNJ—Rio Negro-Juruena, West Africa (LS—Leo Shield). GU—Guri Shear Zone; SAS—Sassandra Shear Zone.

Table 3
Selected paleomagnetic poles.

Continent/formation	Plat (°N)	Plong (°E)	A_{95} (°) dp/dm	Euler pole*	RPlat (°N)	RLong (°E)	Age (Ma)	Ref.
Laurentia				42.86°N, 3.2°E (−48.66°)				
L1—Michikamau intrusives.	02	038	3/7		−02	352	1460 ± 5	1
L2—Harp Lake Complex	−02	026	3/6		−05	350	1450 ± 5	2
L3—St. Francois Mountains	13	039	07		00	009	1476 ± 16	3
L4—Tobacco Root Mountains	−09	036	10		−16	354	1448 ± 49	4
Mean (L1460)	01	035	13		−06	356	1460	
L5—Spokane formation	25	036	05		12	013	1440	5
L6—Purcell Primary	24	036	5		11	012	1443 ± 7	5
L7—Electra Lake Gabbro	21	041	03		06	015	1433 ± 2	6
L8—Laramie anorthosite	07	035	04		−03	002	1429 ± 9	7
L9—McNamara	14	028	07		07	000	1401 ± 6	5
L10—Zig-Zag Dal & Intrusions[#]	−11	049	03		−25	004	1382 ± 2	8
L11—Mackenzie mean	−04	010	05		03	335	1267 ± 2	8
Baltica								
B1—Ropruchey Sills	−41	050	08				1770 ± 12	9
B2—Shosksha Formation	−42	041	07				1790–1770	10
B3—Hoting Gabbro	−43	053	12				1786 ± 10	11
B4—Mean 1780 Ma	−42	048	07					
B5—Components C + D–Sarmatia	−27	169	04	48.0°N,42.2°E (43°)[@]	−40	39	1740–1770	12
B6—Tuna dykes	−21	000	07				1461–1462	13
B7—Salmi Form. vulc. rocks	−06	020	11				1460	13
B8—Lake Ladoga dolerites	−16	357	04				1457 ± 2	13
B9—Valaam monzodioritic – A pole	−14	346	02				1458 ± 4	13
B10—Lake Ladoga mafic rocks	−15	357	06				1452 ± 12	14
Mean (B-1460)	−15	000	13				1460	
B11—Västerbotten-Ulvö-Satakunta[a]	05	339	03				1256–1259	15, 16
B12—Dalarna Complex	−06	340	10				1264–1271	15, 16
B13—Jamtland Complex	01	340	7				1247	15, 15
B14—Mean	00	340	8					
Amazonian Craton				[x] 52.4°N,207.1°E (82.8°)				
A1—Indiavaí Intrusive	−57	250	09		−10	325	1416 ± 7	17
A2—Nova Guarita dykes	−48	246	07		−02	321	1419 ± 4	18
Amazonian Craton				[xx] 70.9°N, 255.6°E (98.8°)				
A3—Colider Intrusive Suite	−63	299	10		−35	34	1789 ± 7	19
A4—Indiavaí Intrusion	−57	250	09		−10	325	1416 ± 7	17
A5—Nova Guarita dykes	−48	246	07		−02	321	1419 ± 4	18
North China Craton				[xx] 54.2°N, 238.2°E (−72.2°)				
NC1—Taihang dykes	−36	067	03		−41	038	1416 ± 7	20,21

Plat, pole latitude; Plong, pole longitude; A_{95}, dp, dm—Fisher's statistic parameters. Ref. 1—Emslie et al. (1976); 2—Irving et al. (1977); 3—Meert and Stuckey (2002); 4—Harlan et al. (2008); 5—Elston et al. (2002); 6—Harlan and Geissman (1998); 7—Harlan et al. (1994); 8—Evans and Mitchell (2011); 9—Fedotova et al. (1999); 10—Pisarevsky and Sokolov (2001); 11—Elming et al. (2009a); 12—Elming et al. (2010); 13—Salminen and Pesonen (2007); 14—Lubnina et al. (2010); 15—Elming and Mattsson (2001); 16—Söderlund et al. (2006); 17—This study; 18—Bispo-Santos et al. (2012); 19—Bispo-Santos et al. (2008); 20—Halls et al. (2000); 21—Piper et al. (2011). * Euler rotation poles used to rotate poles and continents to the Columbia configuration (Fig. 12). RPlat/RPlong—Rotated pole latitude/Rotated pole longitude. [x] Rotation pole used in Fig. 12a. [xx] Rotation pole used in Fig. 12b. # pole from Greenland rotated according to Evans and Mitchell (2011); [a] pole recalculated from Elming and Mattsson (2001). Rotation pole used for West African Craton in Fig. 12a and b, respectively: 6°N, 29°E, −93°, and 16.5°N, 224.3°E, 67.1°. [@] Euler rotation pole used to rotate Sarmatia and its respective pole.

the north and south of North China. Fig. 13 shows an expanded view of Fennoscandia, Volgo-Uralia, Sarmatia, West Africa, Amazonian Craton and North China from Fig. 12b. Based on the ideas of Kusky et al. (2007) we speculate in the presence of a stable triple junction between Fennoscandia, Sarmatia, North China, and Amazonia. Dextral strike slip motion of North China relative to Fennoscandia and sinistral strike slip motion of North China relative to Amazonia/Sarmatia, respectively, occurred after 1780 Ma, and an easternward subduction of oceanic lithosphere developed beneath Sarmatia (Claesson et al., 2001). A break-up of North China is also consistent with the profuse 1780 Ma mafic dyke swarms and sills largely exposed in northern Brazil, Venezuela and Guyana, known as the Avanavero Large Igneous Province (LIP) (Gibbs, 1987; Sial et al., 1987; Choudhuri et al., 1990; Santos et al., 2003). The Avanavero mafic intrusions follow normally the NNW basement structures (Gibbs, 1987). Although most dykes strike NE the Rio Aro dykes in Venezuela strike NW (ca. 1800 Ma, Onstott et al., 1984; Gibbs, 1987), almost parallel with the North China Dykes in the paleogeography of Fig. 13.

Mafic and felsic dykes are also widespread in the Småland Province in southwestern Baltica (Småland-Värmland belt) (Pisarevsky and Bylund, 2010). Some of the mafic dykes strike NW which also nearly parallel the NW dykes from North China in the paleogeography of Fig. 13, while some composite dykes also strike NE (Pisarevsky and Bylund, 2010). Moreover, the 1770 Ma gabbros and dolerites of the Ropruchey sill from eastern Fennoscandia (Fedotova et al., 1999) may represent an extensional regime of the more stable Fennoscandia foreland as reflex of breakup of the North China Craton.

The time Sarmatia rotated into its present position is yet undefined. Elming et al. (2010) suggest it occurred after breakup of Columbia, sometime between 1700 and 1600 Ma with reference to the ages of tectomagmatic peaks in a fault zone adjacent to the NW part of Sarmatia (Bogdanova et al., 2006). However, the consistency of the 1780 Ma and 1470–1420 Ma paleomagnetic poles from Fennoscandia, Laurentia and Amazonian Craton suggests that Sarmatia rotated clockwise thereafter, unless Amazonian and West Africa Cratons have not been solid-linked with Sarmatia, and thus rotated separately. From the age of metamorphism and anatectic melting of deposits along some of the margins of Volgo-Uralia, Sarmatia and Volgo-Uralia are suggested to have collided at ca 2050–2020 Ma ago (Bogdanova et al., 2008). If these cratons welded together at that time it means Sarmatia and Volgo-Uralia together rotated into their present positions relative Fennoscandia. However, there are no paleomagnetic data to test this hypothesis yet.

6. Conclusions

Paleomagnetic analysis of the Indiavaí Intrusive rocks resulted in a well-dated 1415.9 ± 6.9 Ma paleomagnetic pole for Amazonia (IG

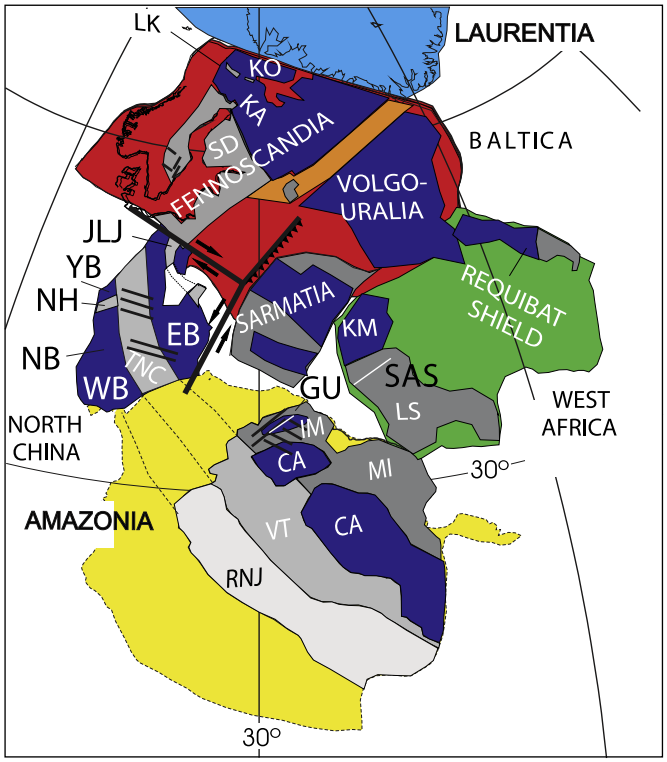


Fig. 13. Expanded view of Fig. 12b. A triple junction between North China, Amazonia/Sarmatia, and Fennoscandia/Volgo-Uralia is shown. Approximate geographic dyke swarm strikes from Amazonia, Fennoscandia, and North China are also shown. Nomenclature as in the legend of Fig. 12 (see text for details).

pole), located at 247.7°E, 57.0°S (N = 16, A_{95} = 8.6°, K = 19.6). Magnetic mineralogy, petrographic analyses and the fact that the host rocks carry stable magnetic components with different directions suggest that the IG pole represents a primary thermo-remanent magnetization, which can be classified with a reliability factor Q = 4. The IG pole is similar to a recent paleomagnetic pole determined for the Nova Guarita mafic dyke swarm for which ^{40}Ar–^{39}Ar dating yield a mean age of 1418 Ma. These new ~1420 Ma poles suggest that the Nova Brasilândia mobile belt, situated between the Nova Guarita and Indiavaí areas, did not represent a suture zone formed during the collision of the Paraguá Terrane and the proto-Amazonian Craton at 1100 Ma ago as previously suggested by Tohver et al. (2004), but rather represent an intracratonic feature. Comparison of these ca. 1420 Ma poles with the Mesoproterozoic poles of Laurentia and Baltica does not support earlier suggested links between southwestern Baltica and northeastern Amazonia at 1460–1410 Ma ago, as based on geological/geochronological data. The 1420 Ma Indiavaí and Nova Guarita poles, however, are consistent with a rotation of Sarmatia (together with Amazonian and West Africa Cratons) as suggested by Elming et al. (2010). This configuration is also consistent with the ca. 1790–1780 Ma reconstructions of Columbia based on paleomagnetic data, where North China may have been located between Baltica and Amazonian Craton (Bispo-Santos et al., 2008). We believe that North China should have drifted away from Columbia, just after its agglutination, which allowed for the collision of Sarmatia with the Fennoscandia/Volgo-Uralia block, ending the formation of Baltica. The time this process happened is yet unknown, and new Proterozoic poles for the Amazonian Craton and Sarmatia (and other intervening blocks) are still needed to completely decipher the Columbia paleogeography and evolution.

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